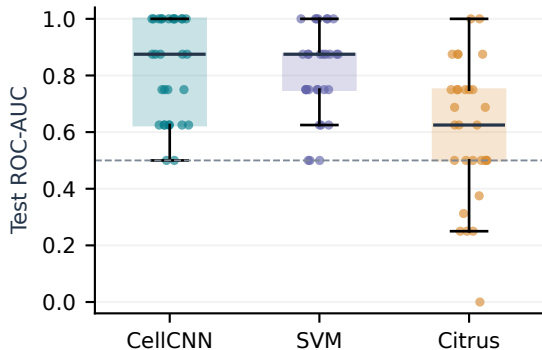
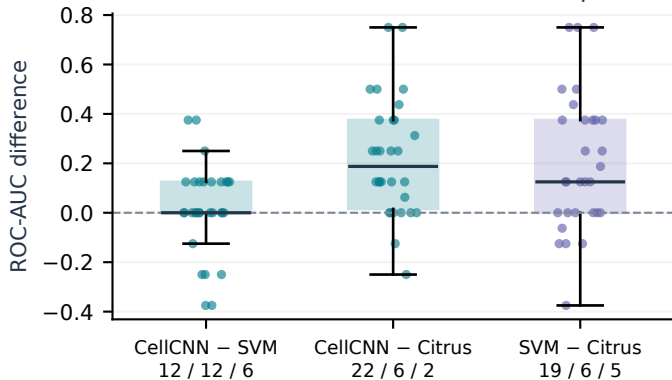


A Classification performance



B Paired differences on the same splits



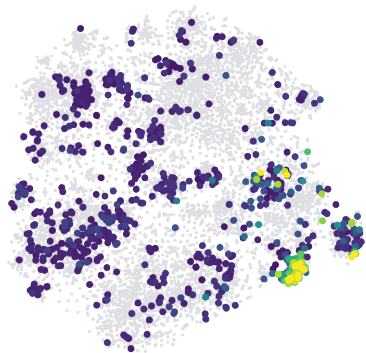
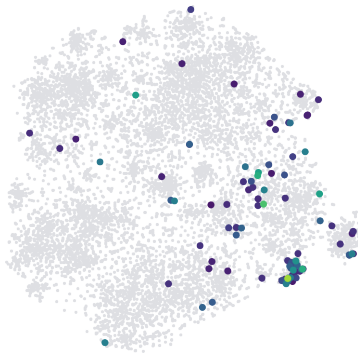
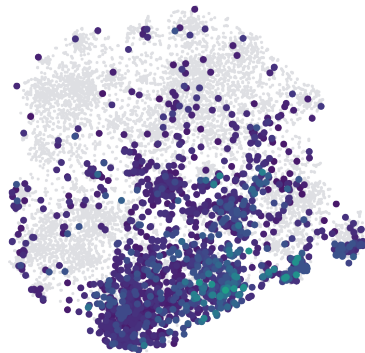
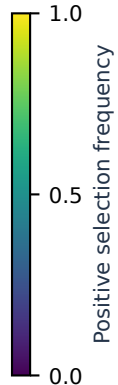
30 shared splits of 20 donors. Boxes: Q1-Q3; line: median; dots: splits.
 Paired counts: better / equal / worse. Split variation is descriptive, not a confidence interval.

A CellCNN

B SVM

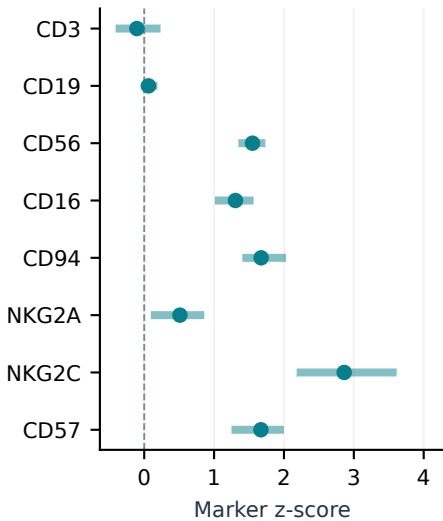
C Citrus

t-SNE 2

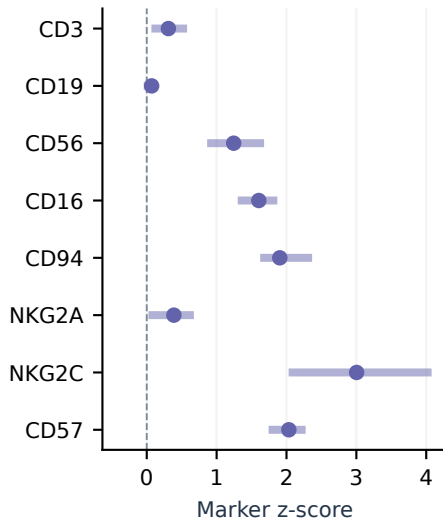
 ≥ 1 visit: 863; $\geq 50\%$: 84 ≥ 1 visit: 93; $\geq 50\%$: 9 ≥ 1 visit: 1583; $\geq 50\%$: 11

All methods: outer test donors only, splits 0-29; same 10,000 reference cells (500/donor).
Denominator: all 4-16 test visits per donor. Grey: never selected. Method-specific selection rules.

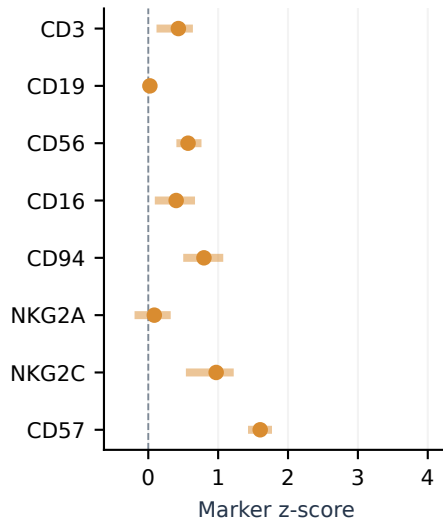
CellCNN: 20 donors



SVM: 20 donors



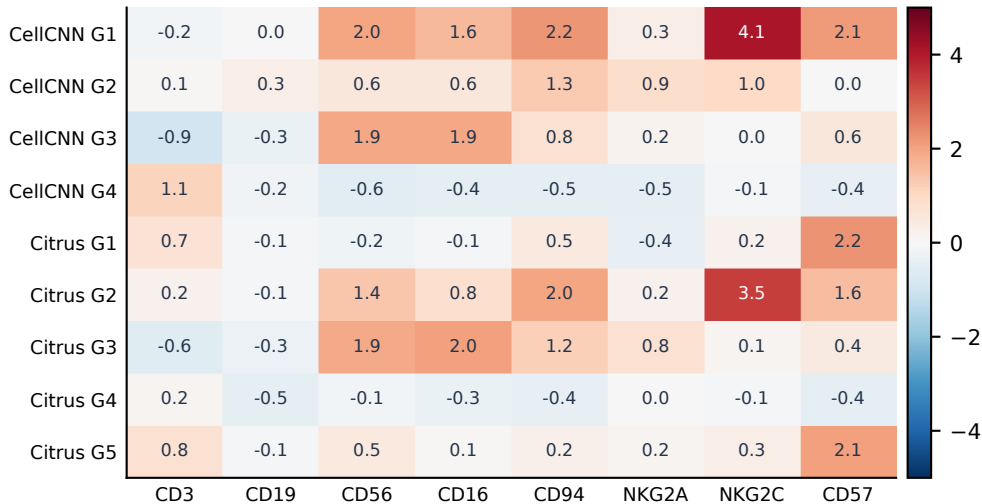
Citrus: 20 donors



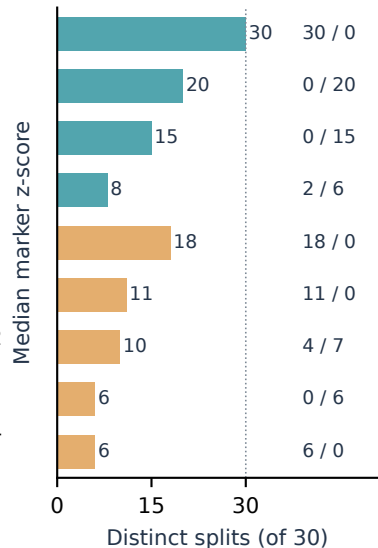
Mean selected cells per nonempty test visit → mean visits per donor → equal donor weights.

Dots: mean across donors; lines: Q1-Q3 across donor profiles, not confidence intervals. Full donor selections; same eight markers.

Marker profiles across recurring groups

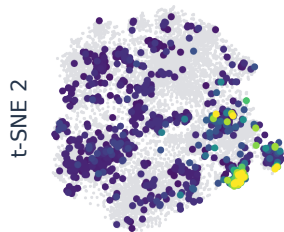


Recurrence | + / -



Groups present in $\geq 6/30$ splits. Average member centroids within each split, then median across splits with that group. Training-subset profiles; + / - : splits with positive / negative members (can overlap). Group absence is not zero expression.

A CellCNN



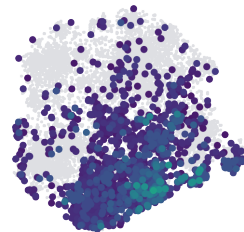
≥ 1 visit: 863; $\geq 50\%$: 84

B SVM

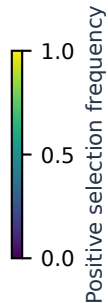


≥ 1 visit: 93; $\geq 50\%$: 9

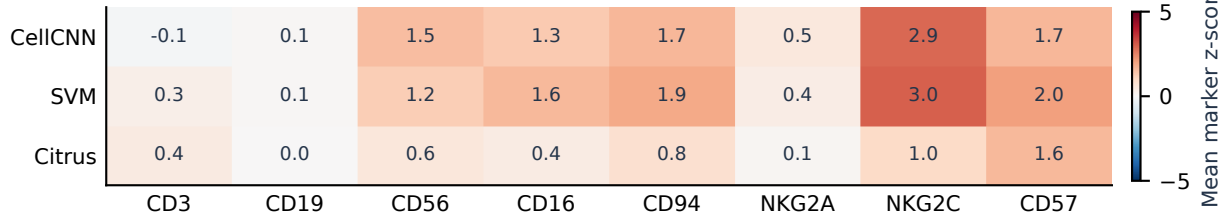
C Citrus



≥ 1 visit: 1583; $\geq 50\%$: 11



D Selected test-cell profiles · equal donor weights



30 shared splits; only held-out donors. Maps: the same 10,000 cells; colour: selection / donor's 4-16 test visits. Profiles: selected full-donor test cells → nonempty visits → equal donor weights. Grey: never selected; rules differ by method.